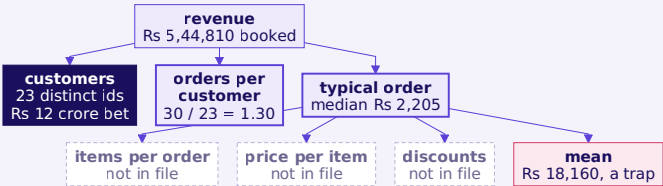


1 The revenue tree, with Monday's numbers on it



Crux. Revenue is customers times orders per customer times average order value, and average order value is items per order times price per item, less discounts.

2 Every branch is a metric with a bill

Branch	Over what	What moving it costs
Customers	A count of distinct ids in the window	Marketing spend
Orders per customer	Distinct customers, same window	Retention
Items per order	Orders	Merchandising
Price per item	Items	Volume, as buyers leave
Discounts	Revenue before discounts	Margin, traded for quantity

The identity checks the tree: $23 \times (30 / 23) \times (\text{Rs } 5,44,810 / 30) = \text{Rs } 5,44,810$.

3 Trap 1, which total is "sales"

Wrong number	Rs 5,44,810 called sales, with 4 cancelled store orders inside it.
Check	Count by status first: 21 delivered, 5 returned, 4 cancelled.
Fix	Not cancelled is Rs 5,35,760 on 26 orders; delivered is Rs 5,20,790 on 21.

Crux. Name the definition before the number: booked, not cancelled, or delivered.

4 Trap 2, rows counted as customers

Wrong number	30 customers, so $30 / 30 = 1.00$ and "nobody comes back".
Check	<code>len(rows)</code> against <code>len(set(ids))</code> : 30 against 23.
Fix	23 customers, 1.30 orders each, 7 came back and 16 bought once.

Crux. Count customers by their id, never by the rows.

5 Trap 3, the mean sold as typical

Wrong number	A first order valued at the mean, Rs 18,160.
Check	Orders above the mean: 1 of 30. Sort and read the top.
Fix	Median Rs 2,205 booked, Rs 2,060 delivered; the top order gets its own line.

Crux. Report the median when one order can move the mean, and say why.

6 Trap 4, lifts added instead of multiplied

Wrong number	Customers +10 percent and frequency +10 percent called 20 percent.
Check	Recompute through the tree: $1.10 \times 1.10 = 1.21$.
Fix	Rs 6,59,220, against Rs 6,53,772 by adding. 15 percent off for 10 percent more quantity is $0.85 \times 1.10 = 0.935$, a 6.5 percent fall.

Crux. Lifts multiply along the tree: two 10 percent lifts make 21 percent.

7 Trap 5, one channel's share

Wrong number	Store brings 91.6 percent of booked revenue, so the plan should be store-led.
Check	Count the orders behind the share; split each channel by status.
Fix	Consumer orders: web Rs 27,290 with 5 of 10 returned, app Rs 18,600 all delivered, store Rs 18,920 with 4 of 9 cancelled.

8 What goes to Meera

"23 customers placed 1.30 orders each at a typical order of Rs 2,205, and 16 bought only once, so open frequency before acquisition; hold the Rs 12 crore until Tuesday's two quarters."

Crux. One window shows the shape of revenue; only two windows show which branch moved.

THE WORDS ON THIS SHEET

- Revenue tree** Revenue drawn as the metrics that multiply into it, less discounts
- Denominator** What a metric is divided by, said before it is computed
- Booked revenue** Every order placed in the window, whatever happened to it later
- Distinct customers** Each customer id counted once, however many orders it placed

- Orders per customer** Orders over distinct customers in the same window
- Median** The middle of the sorted values, or the average of the two middles